Fixed-Kernel Scientific Density-Grid Refinement

Stage 11E-GR3 implementation specification

mdstats

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Contents

Purpose

Stage 11E-GR3 replaces pilot-only hard-coded grid comparisons with one source-bound, fixed-kernel scientific refinement contract. It asks whether a single density hypothesis is numerically stable as only the logical periodic grid is refined.

GR3 does **not** select a bandwidth, alter the Gaussian covariance, choose a rendering backend, simplify a mesh, or admit a browser scene. Those operations belong to other stages. A visual field may be useful without satisfying GR3, and a GR3 certificate may be generated without Plotly or any rendering library.

The implementation owner is:

`mdstats.analysis.density.refinement`

Scope and ownership

GR3 consumes:

- one source/signature context;
- one SAMP0 cross-fit partition signature;
- one fixed positive-definite Cartesian kernel covariance and its signature;
- one scientific resource-policy signature;
- GR0 cell-metric geometry and numerical diagnostics;
- one GR1 exactly nested logical-grid ladder;
- per-level field evidence and, when available, E2 basin and corridor evidence.

GR3 emits orthogonal signed decisions for:

1. scalar density-field resolution;
2. basin-catalog convergence; and
3. corridor/bottleneck convergence.

A converged basin catalog does not imply a converged corridor catalog. Missing or unstable corridor evidence therefore blocks saddle or barrier promotion but does not erase separately converged basin evidence.

Persistent runtime records

The public GR3 records are immutable, canonically serializable, and SHA-256 signed:

```
GridConvergenceStoppingPolicy
ScientificGridRefinementPolicy
DensityFieldLevelEvidence
FeatureGridCorrespondence
BasinGridPairComparison
CorridorGridLevelEvidence
CorridorGridPairComparison
DensityFieldResolutionCertificate
BasinGridConvergenceCertificate
CorridorGridConvergenceCertificate
ScientificGridRefinementBundle
```

Every `from_json_dict` replay validates schema and signature. Tampered payloads fail with `DensityNumericalSerializationError` rather than being accepted as new evidence.

Frozen numerical hypothesis

`ScientificGridRefinementPolicy` binds the complete numerical hypothesis before refinement begins:

```
fixed_kernel_covariance_cartesian
fixed_kernel_signature
scientific_resource_policy_signature
crossfit_partition_signature
coarsest_interval
GridConvergenceStoppingPolicy
FeatureCorrespondencePolicy
metadata
```

The covariance is symmetrized and required to be positive definite. Its smallest standard-deviation scale is

$$\sigma_{\min} = \sqrt{\lambda_{\min}(\Sigma_K)}.$$

A kernel-signature change at any level is a hard input error. Dense and local-sparse field realizations may be compared only when they represent the same source, weights, kernel, logical grid, and normalization semantics. Backend identity is retained as evidence but is not itself a convergence criterion.

`stage11_grid_stopping_v1`

The initial `GridConvergenceStoppingPolicy` preset is identified exactly as:

```
stage11_grid_stopping_v1
```

Its normative defaults are:

Quantity	Default
Grid refinement factor	2
Physical-resolution gate	$\Delta_{\max}/\sigma_{\min} \leq 0.5$
Maximum number of levels	8
Consecutive passing level pairs	2
Maximum probability-field L^1 change	0.02
Maximum normalization residual	10^{-6}
Basin count	unchanged
Maximum basin-anchor motion	$0.10\sigma_{\min}$
Minimum matched-basin overlap	0.95
Maximum basin-probability change	0.02
Basin ambiguity/split/merge/unmatched	prohibited
Corridor adjacency	unchanged
Minimum matched-corridor overlap	0.90
Maximum bottleneck motion	$0.15\sigma_{\min}$
Maximum relative width change	0.10
Maximum relative bottleneck-density change	0.10
Corridor ambiguity/split/merge	prohibited

These are versioned conservative defaults, not universal physical constants. Any altered preset must receive a new policy identifier and regression fixtures. An explicit custom policy retains all resolved numerical values in its signed certificate.

Deterministic nested ladder

`plan_scientific_grid_refinement` delegates logical-grid construction to GR1. For the initial preset, each level refines every grid axis by exactly a factor of two. The realized Cartesian interval is computed in the full cell metric.

The minimum physical-resolution interval is

$$\Delta_{\text{physical}} = 0.5\sigma_{\min}.$$

Because GR3 requires **two consecutive** passing level-pair comparisons after that gate, the planner requests an additional post-gate level. With refinement factor r and m required consecutive comparisons,

$$\Delta_{\text{requested}} = \frac{\Delta_{\text{physical}}}{r^{m-1}}.$$

For `stage11_grid_stopping_v1`, this is $\Delta_{\text{requested}} = 0.25\sigma_{\min}$. This is not a stronger physical-resolution assertion; it merely ensures that two eligible comparisons can exist.

The ladder preserves GR1 status:

```
target_reached
budget_limited
level_limited
```

A finest affordable level is retained as diagnostic evidence but cannot be silently promoted to convergence.

Field-level evidence

`DensityFieldLevelEvidence` binds each logical-grid level to:

```
level index and grid shape
realized Cartesian intervals
field-estimate signature
fixed-kernel signature
backend label
probability and number normalization residuals
probability-field L1 and optional Linf changes from the previous level
CIC covariance
periodic Gaussian-stencil covariance
effective artificial-broadening covariance
metadata
```

For direct periodized-Gaussian estimators that do not use CIC deposition or a separate discrete convolution, CIC and stencil covariance are explicitly zero. Nested-field comparisons restrict the fine field to the exactly corresponding coarse nodes; interpolation is not introduced as an unrecorded hypothesis.

A field pair is eligible only when the fine level satisfies $\Delta_{\max}/\sigma_{\min} \leq 0.5$. It passes when:

- both coarse and fine normalization residuals do not exceed 10^{-6}; and
- the probability-field L^1 change does not exceed 0.02.

`DensityFieldResolutionCertificate` is `converged` only when the tail of the eligible ladder contains at least two consecutive passing pairs. The certificate retains pair eligibility, pair pass/fail values, the accepted level, and reason codes.

Basin correspondence and convergence

Basin comparisons use the signed SAMP0 `stage11_feature_correspondence_v1` policy. Its normalized assignment cost is

$$C_{ij} = 1\left(\frac{d_{ij}}{\sigma_{\min}}\right)^2 + 2(1 - O_{ij}) + 1\frac{|p_i - p_j|}{p_{\mathrm{scale}}}.$$

The initial correspondence preset uses:

```
maximum assignment cost = 3.0
ambiguity margin = 0.10
deterministic tie break = lexicographic_feature_id
admissible types = point-to-point and ridge-to-ridge
explicit outcomes = matched, ambiguous, unmatched, split, merge
```

`compare_basin_catalog_pair` evaluates periodic anchor distance in the exact cell metric, overlap of nested owner maps, integrated probability change, candidate type, assignment cost, ambiguity, unmatched features, and positive- overlap split/merge alternatives. Unsupported nodes remain outside overlap claims.

A basin pair passes only when all `stage11_grid_stopping_v1` basin tolerances hold. `BasinGridConvergenceCertificate` then applies the same physical-resolution eligibility and two-consecutive-pair rule as the field certificate.

Corridor evidence and convergence

`CorridorGridLevelEvidence` records, per accepted basin pair:

```
adjacency identity
periodic bottleneck position
corridor width
bottleneck density
optional support-node identity
catalog signature
```

`compare_corridor_level_pair` records adjacency equality, minimum support intersection-over-union when support nodes are supplied, periodic bottleneck motion, maximum relative width and density changes, split/merge records, ambiguity, and evidence completeness.

A corridor pair passes only when the initial corridor thresholds hold. Missing width, bottleneck, density, or support evidence is not converted into a zero or a passing value. It yields an unresolved corridor certificate with the reason:

```
corridor_width_or_support_evidence_unavailable
```

Thus the valid combined outcome

```
field numerics: converged
basins: converged
corridors: unresolved_due_to_missing_evidence
```

is preserved rather than collapsed into one Boolean.

Convergence statuses

GR3 uses the exact `GridConvergenceStatus` values:

```
converged
unresolved_due_to_resolution_budget
unresolved_due_to_refinement_limit
unresolved_due_to_insufficient_passing_levels
unresolved_due_to_metric_failure
unresolved_due_to_missing_evidence
```

The distinction is normative:

- `unresolved_due_to_resolution_budget` means GR1 could not allocate the requested ladder under the signed scientific resource policy;
- `unresolved_due_to_refinement_limit` means the configured level depth was exhausted;
- `unresolved_due_to_insufficient_passing_levels` means too few eligible passing comparisons exist despite a valid ladder;
- `unresolved_due_to_metric_failure` means one or more required comparisons failed their numerical or topology tolerances; and
- `unresolved_due_to_missing_evidence` means the relevant field, basin, or corridor inputs were not supplied completely.

These are scientific outcomes, not substitutes for exceptions caused by invalid cells, malformed records, signature mismatch, or a kernel change.

Source, resource, and cross-fit binding

The bundle binds:

```
source bundle signature
scientific refinement-policy signature
scientific resource-policy signature
cross-fit partition signature
GR1 ladder signature
field certificate signature
basin certificate signature
corridor certificate signature
```

A caller-supplied resource policy must match the policy's signed resource identity before planning. Evidence from another source or cross-fit partition must not be pooled implicitly.

GR3 operates only within the discovery/model-selection domains authorized by SAMP0. Held-out basin or corridor validation cannot retune the kernel, ladder, candidate count, correspondence policy, or accepted grid. Final candidate selection and freezing remain Stage 11E-GR4 responsibilities.

Integration with E1 and E2

E1 provides fixed-kernel density realizations and normalization/numerical metrics for the GR1 ladder. E2 provides deterministic per-level basin and density-boundary candidates. GR3 certifies only numerical grid stability.

GR3 does not replace:

- bandwidth/model selection;
- SAMP1 basin recurrence and effective-sample adequacy;

- SAMP2 passage/corridor support;
- STAT2/STAT3 thermodynamic admissibility;
- E3 force or mean-force evidence; or
- GR4 final hypothesis freeze.

Failure behavior

Fail closed for:

- a nonfinite or singular cell;
- a non-positive-definite kernel covariance;
- malformed or non-SHA-256 source, kernel, resource, or partition signatures;
- a kernel change inside one ladder;
- non-nested or level-misaligned evidence;
- source/resource/cross-fit identity mismatch;
- corrupted canonical JSON or signature mismatch;
- a rendering/browser budget presented as a scientific resource policy; or
- held-out evidence used to tune the numerical hypothesis.

A valid but unconverged ladder returns a signed unresolved certificate instead of raising an exception.

Acceptance tests

The focused GR3 boundary requires:

- exact default values and policy replay;
- one additional post-gate ladder level for the two-comparison rule;
- two consecutive eligible passing field comparisons;
- diagnostic-only treatment of one passing pair under a budget limit;
- metric-failure rejection without finest-level promotion;
- fail-closed kernel change;
- periodic basin correspondence with ambiguity, split, merge, and unmatched handling;
- independent basin and corridor convergence;
- unresolved corridor evidence without deleting basin convergence;
- source/resource/cross-fit signature rejection;
- canonical bundle replay and tamper rejection;
- public API and documentation-contract tests; and
- import isolation from plotting and rendering policy.

The broader compatibility boundary includes GR0–GR2, density estimators, atomic/framework density, E0a–E8a structural contracts, and clean-wheel replay.

Implementation status and next stage

Implemented in `0.20.26a0` under architecture revision 56. The implementation adds the common fixed-kernel scientific refinement runtime without changing the GR2 visual-policy outputs or promoting the historical Stage 11E8a pilot.

The next implementation stage is **Stage 11E-GR4: cross-fitted numerical- hypothesis selection and freeze**.